

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 1

Kopiereg voorbehou

Module EBN111
Elektrisiteit en Elektronika
15 Maart 2012

Semester Test 1

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Module EBN111
Electricity and Electronics
15 March 2012



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: Exam information:

Maksimum punte: Maximum marks:	45	Volpunte: Full marks:	45
Duur van vraestel: Duration of paper:	90 minute 90 minutes	Oopboek / toeboek: Open / closed book:	Toe Closed
Totale aantal bladsye (hierdie blad ingesluit): Total number of pages (including this page):	13		

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.
Final answers must be written as real numbers, and not as fractions.

Dosent(e): Lecturer(s):	1 J Joubert
	2 JD le Roux

Groep hoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding.
Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.

Groep hoof: Group Head:	Prof J Joubert	Handtekening: Signature:	
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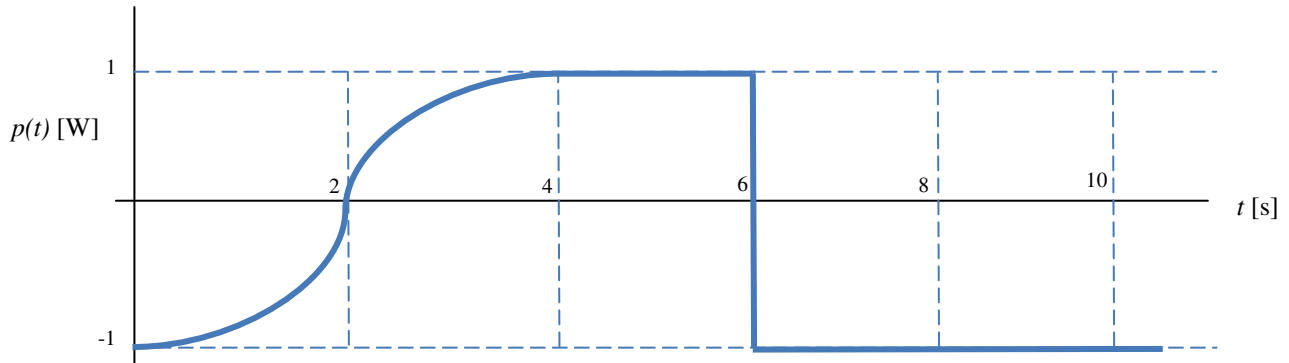
VRAAG/QUESTION 1 [22]

Vraag 1.1 [2]

Die grafiek dui die drywing van 'n element teenoor tyd aan. Bepaal die hoeveelheid energie wat die element verbruik tussen 4 en 10 sekondes.

Question 1.1 [2]

The graph shows the power of an element over time. Determine the energy used by the element between 4 and 10 seconds.



Vraag 1.2 [2]

Dit verg 600mJ om -60 nC in 6 μ s deur 'n element te beweeg. Bepaal die spanning oor die element en die stroom deur die element. (Skryf die antwoorde deur gebruik te maak van die SI voorvoegsels: M, k, m, μ , n, of p.)

Question 1.2 [2]

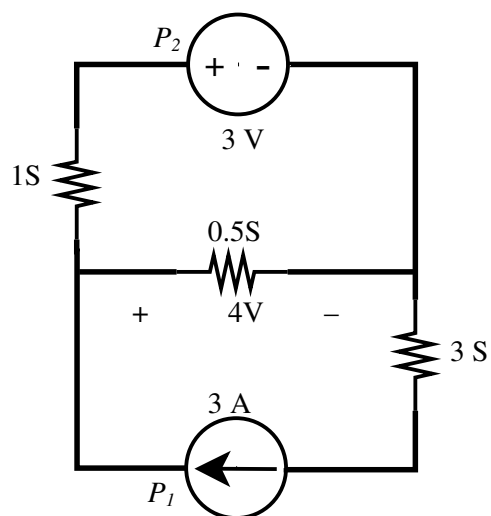
To move -60 nC in 6 μ s through an element requires 600mJ. Determine the voltage across the element and the current through the element. (Write the answers using the SI prefixes: M, k, m, μ , n, or p.)

Vraag 1.3 [3]

- a) Bepaal die drywings P_1 en P_2 .
b) Vir P_2 , word drywing geabsorbeer of gelewer?

Question 1.3 [3]

- a) Determine the power P_1 and P_2 .
b) For P_2 , is power absorbed or provided?



Vraag 1.4 [7]

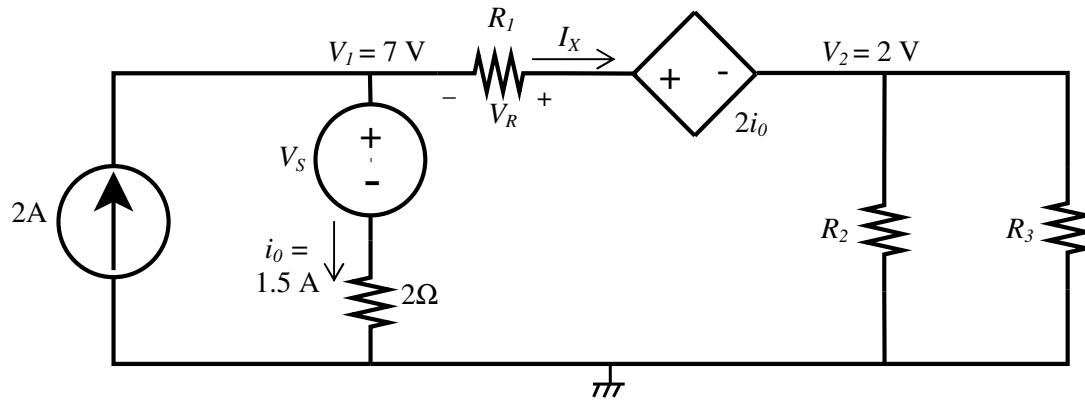
Weerstande R_1 , R_2 en R_3 en bronspanning V_S is onbekend.

- (a) Hoeveel nodusse is daar in die stroombaan? [1]
- (b) Hoeveel takke is daar in die stroombaan? [1]
- (c) Bepaal die spanning V_S . [1]
- (d) Bepaal die spanning V_R oor weerstand R_1 . [2]
- (e) Bepaal stroom I_X . [2]

Question 1.4 [7]

Source voltage V_S and resistors R_1 , R_2 and R_3 are unknown.

- (a) How many nodes are there in the circuit? [1]
- (b) How many branches are there in the circuit? [1]
- (c) Determine the voltage V_S . [1]
- (d) Determine the voltage V_R over resistor R_1 . [2]
- (e) Determine the current I_X . [2]

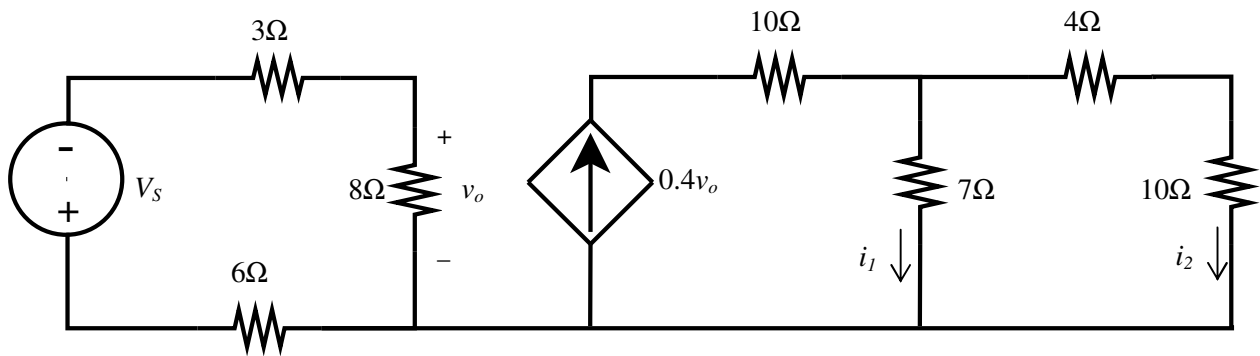


Vraag 1.5[4]

- a) Bereken v_o indien $V_S = 21.25$ V.
b) Bereken i_2 indien $v_o = 5$ V.

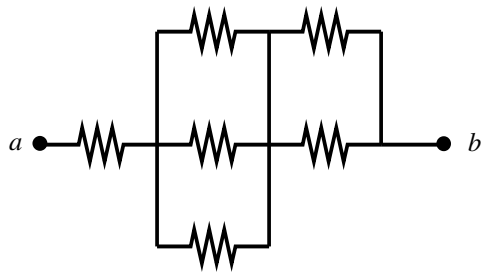
Question 1.5[4]

- a) Determine v_o if $V_S = 21.25$ V.
b) Determine i_2 if $v_o = 5$ V.

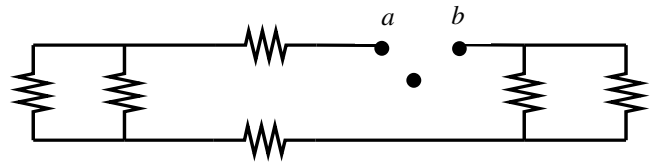


Vraag 1.6[4]

Bepaal die ekwivalente weerstand by terminale a - b vir stroombaan (1) en (2). Alle weerstande is $2\ \Omega$.

Stroombaan/Circuit (1)**Question 1.6[4]**

Determine the equivalent resistance at terminals a - b for circuit (1) and (2). All resistors are $2\ \Omega$.

Stroombaan/Circuit (2)

VRAAG/QUESTION 2 [23]

Vraag 2.1[4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan:

$$aV_1 + bV_2 = -200$$

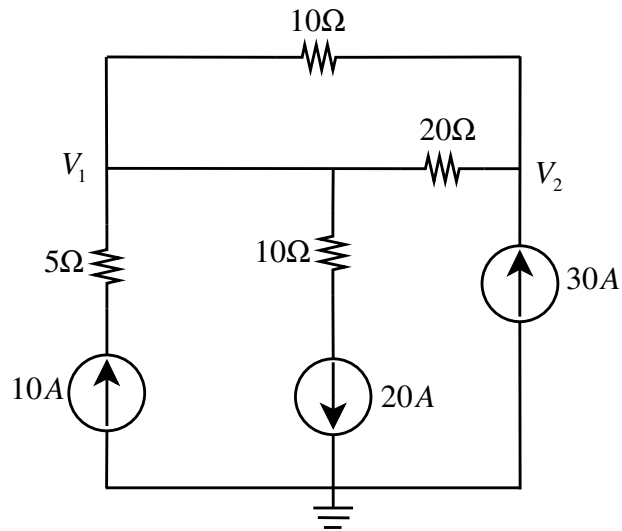
$$cV_1 + dV_2 = 600$$

Question 2.1[4]

Use node analysis to set up two equations of the following form for the circuit:

$$aV_1 + bV_2 = -200$$

$$cV_1 + dV_2 = 600$$



Vraag 2.2[4]

Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan:

$$ai_1 + bi_2 = 0$$

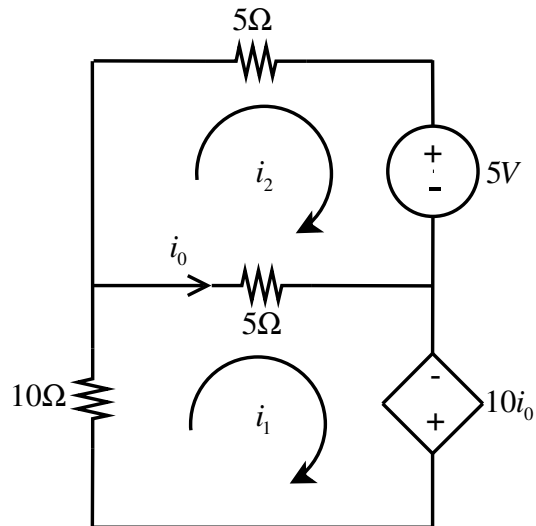
$$ci_1 + di_2 = -5$$

Question 2.2 [4]

Use mesh analysis to set up two equations of the following form for the circuit:

$$ai_1 + bi_2 = 0$$

$$ci_1 + di_2 = -5$$



Vraag 2.3 [4](a) Bepaal V_I .

$$\begin{bmatrix} 5 & -3 \\ -3 & 6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

(b) Bepaal i_2 .

$$\begin{bmatrix} 5 & -5 & 0 \\ -2 & 20 & 0 \\ 0 & 0 & 10 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \\ -10 \end{bmatrix}$$

Question 2.3 [4](a) Determine V_I .

$$\begin{bmatrix} 5 & -3 \\ -3 & 6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

(b) Determine i_2 .

$$\begin{bmatrix} 5 & -5 & 0 \\ -2 & 20 & 0 \\ 0 & 0 & 10 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \\ -10 \end{bmatrix}$$

Vraag 2.4 [4]

Gebruik node-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan:

$$aV_1 + bV_2 = 10$$

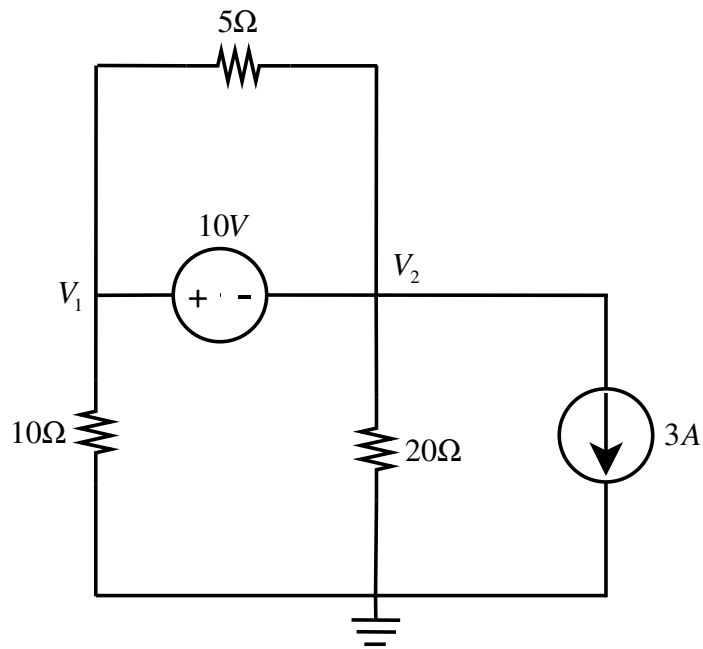
$$cV_1 + dV_2 = -60$$

Question 2.4 [4]

Use node analysis to set up two equations of the following form for the circuit:

$$aV_1 + bV_2 = 10$$

$$cV_1 + dV_2 = -60$$



Vraag 2.5 [4]

Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan:

$$ai_1 + bi_2 = 10$$

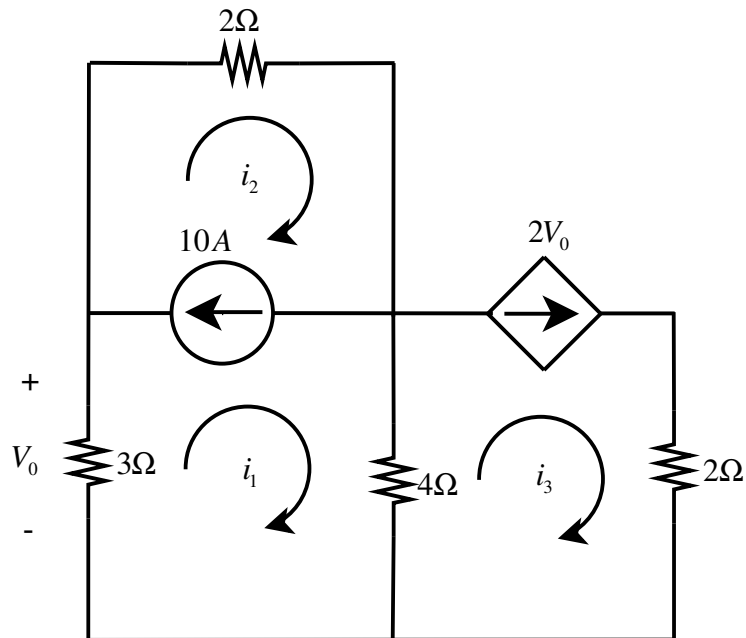
$$ci_1 + di_2 = 0$$

Question 2.5 [4]

Use mesh analysis to set up two equations of the following form for the circuit:

$$ai_1 + bi_2 = 10$$

$$ci_1 + di_2 = 0$$

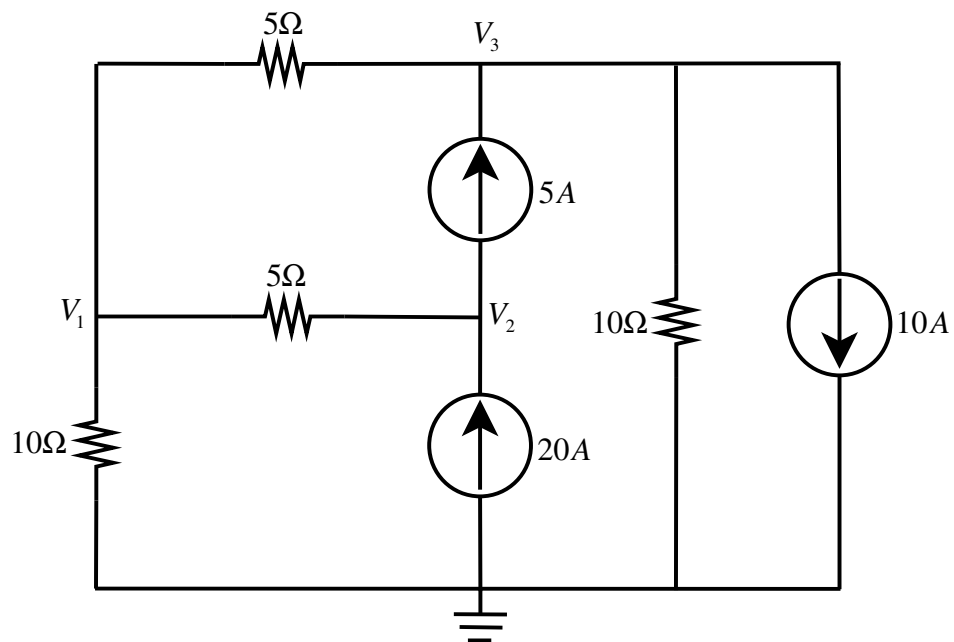


Vraag 2.6 [3]

Bepaal die node-spannings vergelykings deur inspeksie.

Question 2.6 [3]

Determine the node-voltage equations by inspection.



EBN111 SEMESTER TOETS/TEST 1 (15 MAART 2012)
ANTWOORDBLAD / ANSWER SHEET

Student se besonderhede/Student's details									
Van en voorletters: <i>Surname and initials:</i>		Studentenommer: <i>Student number:</i>							
Graadkursus: <i>Degree Course:</i>									
Tel. nr.: <i>Tel no.:</i>									

Q1		Q2		Tot:	/45
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VRAAG 1 – QUESTION 1 (22)		
1.1	$W =$	
1.2	$v =$	$i =$
1.3	$a) P_1 =$ $P_2 =$	
	$b)$	
1.4	$a) \text{ Nodes:}$	$b) \text{ Branches:}$
	$c) V_S =$	$d) V_R =$
	$e) I_X =$	
1.5	$a) v_0 =$	$b) i_2 =$
1.6	<i>Circuit (1):</i> $R_{eq} =$	<i>Circuit (2):</i> $R_{eq} =$